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Determinants of students' adoption of AI chatbots in higher education: the moderating role of tech readiness

Muhammad Khalilur Rahman
Faculty of Entrepreneurship and Business, Universiti Malaysia Kelantan, Kota Bharu, Malaysia

Md Arafat Hossain
Southeast Business School, Southeast University, Dhaka, Bangladesh

Noor Azizi Ismail
Malaysian Graduate School of Entrepreneurship and Business, Universiti Malaysia Kelantan, Kota Bharu, Malaysia

Mohammad Shahadat Hossen
Faculty of Applied Social Sciences, Universiti Sultan Zainal Abidin, Kuala Terengganu, Malaysia, and

Moniya Sultana
Department of Global Business, Kyungsung University, Busan, Republic of Korea

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Abstract

Purpose – This study aims to investigate the key factors influencing students' adoption of artificial intelligence (AI) chatbot applications in higher education. It further examines the mediating and moderating role of AI chatbots and tech readiness in determining the effect of perceived usefulness, subjective norms, tech simplicity and tech literacy on the intention to use AI chatbot applications.

Design/methodology/approach – A survey was conducted at Malaysian universities whereby 430 students participated and 426 responses were deemed valid for analysis. The data was carefully examined to ensure the accuracy of the proposed model. To comprehend the intricate relationships in the model, the authors used the partial least squares structural equation modeling (PLS-SEM) technique.

Findings – The results revealed that perceived usefulness (PU), subjective norm (SN), tech literacy (TL) and tech simplicity (TS) have a significant impact on students' intention to use AI applications. The intention to use AI chatbot applications significantly influences the adoption of AI chatbots in higher education. The findings indicated that the intention to use AI chatbots mediates the effect of PU, SN, TL and TS on the adoption of AI chatbots. Tech readiness (TR) moderates the effect of PU and TS on the intention to use AI chatbot applications.

Originality/value – This research addresses a new insight into AI chatbot adoption within higher education, particularly demonstrating how AI chatbots and tech readiness act as mediators and moderators in shaping students' perceptions and adoption of AI chatbot applications.

Keywords AI chatbots, Subjective norm, Tech literacy, Tech readiness, Higher education

Paper type Research paper

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1. Introduction

Higher education is rapidly changing due to the emergence of new learning technologies like artificial intelligence (AI) chatbots, which provide learners with immediate feedback, custom learning assistance and automated answers to students' questions. AI chatbots are considered one of the more effective and simple ways of learning in a digital environment because they increase engagement and learning effectiveness (Han *et al.*, 2025; Oydna *et al.*, 2024; Hu, 2024). Furthermore, many universities have developed chatbot applications within their learning management systems to help students with academic, administrative and real-time learning tasks. Nevertheless, there are some concerns over the effectiveness of AI chatbots as the perception of students, their technological preparedness and external factors differ (Chen *et al.*, 2025; Hopkyns, 2022; Dahri *et al.*, 2024). The use of AI chatbots for learning in higher education has become increasingly popular.

Several studies have dealt with issues such as adoption intention, perceived usefulness and perceived ease of use concerning the adoption of computers, e-commerce, mobile learning, massive open online courses (MOOCs) and virtual learning environments (Sumi *et al.*, 2025; Al Oraini, 2024; Gurban and Almogren, 2022; Gupta, 2020; Ma and Lee, 2019). Although these investigations help to understand the adoption of technology in general, there is still an important gap concerning how perceived usefulness, simplicity of the technology, literacy and social or subjective norms affect students' intention to use AI chatbot applications in higher education (Polyportis and Pahos, 2025; Chanda *et al.*, 2025; Foroughi *et al.*, 2025).

Many studies have investigated the technological functions of AI chatbots in education, such as providing personalized learning experiences, offering real-time assistance and giving automated feedback (Casheekar *et al.*, 2024; Lin and Yu, 2024). However, there are gaps in the behavioral aspects of students to their perceptions and social determinants of willingness to use AI chatbot tools for distance learning. The impact of subjective norms, or the impact of colleagues, teachers and institutional support, is still not well understood when it comes to AI chatbot use in higher education. In addition, there is insufficient research regarding the impact of technological literacy, and students' basic ability to use AI chatbot applications for academic purposes in higher education.

The perceived usefulness and technological simplicity are important factors in the adoption decisions of students for learning effectiveness. For example, if students see chatbot applications as beneficial to their studies and easy to use, they would be more inclined to use them in their education (Foroughi *et al.*, 2025; Guo and Li, 2024). Technological literacy is also an important consideration. More digitally literate students are expected to adopt AI-based learning tools with ease, while those with low digital literacy may face hurdles in adoption (Ait Baha *et al.*, 2024). Positive comments by instructors, peer students and institutional policy also represent a form of subjective norm that affects the students' intention to use AI chatbots. However, this body of work has not addressed the issue of social influence and normative social information related to the use of AI chatbots (Demyanova, 2024). Moreover, a student's willingness and preparedness to accept the use of AI-based learning tools could affect the relationship between these factors and the adoption of AI chatbots. While prior studies have investigated the effectiveness of AI chatbots, limited attention has been given to how students' technological readiness influences their adoption behavior (Casheekar *et al.*, 2024; Lin and Yu, 2024). To address these gaps, this study aims to investigate the following research questions:

RQ1. What are the key factors influencing students' intention to adopt AI chatbot applications in higher education?

RQ2. How does technological readiness moderate the relationship between perceived usefulness, subjective norms, technological simplicity, literacy and AI chatbot adoption?

This research aims to analyze the main determinants that influence higher education students' use of AI chatbots for education purposes and assess how technological readiness modifies the effect of perceived usefulness, subjective norms, simplicity of technology and literacy on AI chatbot adoption. In addressing these issues and research objectives, the study aims to provide novel theoretical contributions concerning the adoption of AI chatbots in higher education, as well as practical contributions to universities, educators and technology developers. Evaluating the determinants of chatbot adoption is fundamental to building sophisticated learning systems geared toward the students' needs and technological abilities. The remaining parts will address pertinent literature, the technology acceptance model (TAM), the unified theory of acceptance and use of technology (UTAUT), explain the research methods and provide data to achieve the aims of the work.

2. Literature review

2.1 Underpinning theory

This study used the concept of the Technology Acceptance Model (TAM) (Marangunić and Granić, 2015) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh and Zhang (2010) to evaluate students' adoption of AI chatbot application in higher education. According to TAM, the two key elements that influence an individual's intention to use technology are perceived usefulness and perceived ease of use (Szymkowiak and Jeganathan, 2022; Rahman *et al.*, 2021). The TAM describes an individual's decision to adapt to new technology (Teo, 2010). Perceived usefulness affects adoption as students are willing to use AI chatbots if these tools aid in achieving their learning goals (Teo, 2010). Some students perceive online learning to be an efficient method for them to access course materials, interact with peers and receive constructive feedback from teachers (Du *et al.*, 2025).

In higher education, subjective norms play a crucial role in students' adoption of AI chatbots for learning (Polyportis and Pahos, 2025). The TAM and UTAUT emphasize social influence in the adoption of new technologies. When students believe that their peers, teachers or organizations expect them to use AI chatbots, they are more willing to use these tools for learning. Technology literacy refers to students' self-efficacy and skill in the use of AI chatbots (Rahman *et al.*, 2023; Ji *et al.*, 2025). Based on TAM, perceived ease of use sets a student's intention toward technology adoption, thus implying users with higher digital literacy will find AI chatbots easier to use and more suitable for learning purposes (Romero *et al.*, 2020). Similarly, UTAUT focuses on performance expectancy which states that people adopt technology for positive reasons believed to improve their educational experiences. Technological literacy is necessary for adoption.

Students with advanced digital skills are more likely to adopt the technology, while those with low skills may have adoption barriers (Ait Baha *et al.*, 2024). Greater technological proficiency enhances one's self-efficacy regarding AI learning tools, and students tend to use those tools more frequently (Romero *et al.*, 2020). Technology simplicity also has an impact on adoption. Students who believe that AI-based chatbots are effortless to use are more likely to interact with them in the learning process (Foroughi *et al.*, 2025). When students have positive experiences with chatbots, they are likely to use them more often in their study practices. In addition, tech readiness acts as a moderator for the impacts of perceived usefulness and tech simplicity on students' adoption intention of AI chatbots. The students have a positive intention toward the new technology and seem to be ready to accept the

advantages and simplification offered by the AI chatbots resulting in increasing adoption (Casheekar *et al.*, 2024; Lin and Yu, 2024).

2.2 Perceived usefulness

Perceived usefulness (PU) is concerned with an individual's beliefs relating to any given technology improving their work or learning processes. PU is one of the most important determinants in the TAM (Davis, 1989) and has been extensively analyzed in the field of technology adoption. As for AI chatbots, PU impacts both user acceptance and continued usage. AI-powered chatbots with advanced functionalities such as natural language processing and machine learning along with tailored services can yield quicker responses and enhance the learning experience (Lorenzo *et al.*, 2025). Students are more inclined to use AI chatbots for learning when they perceive them to be completing tasks rapidly and providing appropriate support (Monib *et al.*, 2025). According to Al-Shafei (2025), users become more satisfied and engaged with chatbots that respond accurately and in a relevant manner. Zhou *et al.* (2025) argue that chatbot communication is facilitated by the development of AI technology, thus increasing users' positive confidence in the system. In the context of higher education, perceived usefulness (PU) reflects students' appreciation of AI chatbots as valuable learning resources. Chatbot adoption is more likely when students perceive their positive learning experience to be improved using chatbots. AI chatbots are supportive and emotive tools, especially in the roles of personal task supporter and feedback provider (Al-Shafei, 2025). This positive perception of usefulness aids in chatbot application engagement. Perceived usefulness has a positive and significant effect on students' behavioral intention to use AI applications (Awal and Haque, 2025). Therefore, young learners are likely to continue using AI applications when they find them useful and are satisfied with their performance. Based on this premise, this research proposes that:

H1. Perceived usefulness has a significant impact on students' intention to use AI chatbot applications.

2.3 Subjective norms

Subjective norms, or social influence, notably impact students' intentions toward employing AI chatbot applications. If friends, colleagues or others depend on these tools, there is a greater chance that students will use these tools. Ma (2025) reported that subjective norms predicted students' intention to use AI tools in learning. Al-Qaysi *et al.* (2024) focused on students learning English as a second language and confirmed the positive impact of subjective norms on the intention of the learners to use the chatbot for learning purposes. Pramod *et al.* (2025) examined how subjective norms affect consumers' evaluations of chatbots in customer service environments. The usage of ChatGPT among university students found that social norms within educational settings seem to have a positive impact on the utilization of AI chatbots (Stohr *et al.*, 2024). Factors predicting the intensity of AI chatbot interactions among students' behavioral intentions were significantly predicted by subjective norms (Islam *et al.*, 2024). The field of study and the features of the chatbot applications may influence the impact of subjective norms toward the intention to use AI chatbot applications. As a result, this study suggested that:

H2. Subjective norm has a significant impact on students' intention to use AI chatbot applications.

2.4 Tech literacy

Technological literacy encompasses the skillful use, management and knowledge of technology and is becoming vital in understanding students' willingness to use AI chatbots in a learning environment (Hur, 2025; Wu *et al.*, 2025). Students with higher technological literacy are better able to incorporate AI into their learning activities, which helps to increase productivity, engagement and overall interest in class activities (Southworth *et al.*, 2023; Singh *et al.*, 2025). The research observed that, with the aid of AI, students can read and write with greater ease and actively participate in more advanced thinking and dialogue (Kim *et al.*, 2025). The rapid adoption of AI technologies could develop concerns regarding over-dependence that might impact critical thinking and problem-solving. Singh *et al.* (2025) pointed out how universities are going back to their traditional exam formats. Those students with higher degrees of technological literacy are more likely to regard the AI chatbot applications as useful tools that can assist in their learning. Such perception increases their willingness to use the technologies. For example, Şahin Kızıl *et al.* (2025) investigated the adoption of AI chatbots among students and identified that technological literacy impacted their perception of the usefulness of the chatbots, which subsequently positively increased their intention to use AI chatbot applications. Therefore, we postulated that:

H3. Tech literacy has a significant impact on the intention to use AI chatbot applications.

2.5 Tech simplicity

Technological simplicity referred to as perceived ease of use is a significant element that impacts users' willingness to accept new technologies (Hussain *et al.*, 2025; Pagnanelli *et al.*, 2025). Within the context of the TAM, perceived ease of use of tech simplicity is described as the belief that using a system would require minimal effort. This phenomenon has received a lot of attention in the literature for AI chatbot applications, among other technologies. Several studies proved the positive correlation between the ease of technology and the willingness to use AI chatbots (Alotaibi and Rehman, 2025). For example, Foroughi *et al.* (2025) stated that perceived ease of use has a significant positive impact on users' behavioral intentions toward interacting with AI-based chatbots. Ma *et al.* (2025) focused on the acceptance of AI applications and confirmed that ease of use has a positive impact on users' intention which subsequently affects adopting advanced technologies. Students are more inclined to use learning tools that integrate AI features, meaning that the technology's ease of use is critical for its adoption. Moreover, Alotaibi and Rehman (2025) showed that positive user attitudes toward AI chatbots increase their easy usage, which supports the users' online learning experiences. Bahaddad (2025) suggested that user-friendly AI chatbots need to be designed with a personal touch and adaptable educational components to keep students interested throughout the learning process. Despite these findings, certain studies claim that perceived ease of use does not affect the intention of employing AI chatbots. Hussain *et al.* (2025) reported that perceived usefulness has a positive influence on users' intentions, but perceived ease of use did not have a significant impact. Based on this premise, this research proposes that:

H4. Tech simplicity has a significant impact on the intention to use AI chatbot applications.

2.6 Use of AI chatbots intention

The actual adoption of AI chatbots in higher education learning depends on students' intention to use them, which is influenced by factors such as perceived usefulness, literacy,

tech simplicity and social influence. Using the TAM concept (Davis, 1989), and the UTAUT (Venkatesh *et al.*, 2003), intention serves as a dimension that has a strong influence over technology adoption. In the case of online classes, learners are willing to use the AI chatbot and enhance their academic engagement and effectiveness. Many studies highlight the role of intention in the adoption of AI chatbots (Awal and Haque, 2025). For example, Saha *et al.* (2025) reported that students' intention to use learning AI tools strongly predicts their actual adoption, thus enriching their academic experience. Awal and Haque (2025) noticed that in case students have a positive perception of AI chatbots and view them as user-friendly, their intention to use these tools increases, fostering further dependence on them for learning. Wang *et al.* (2023) also verified that a strong intention to use AI chatbots is associated with increased usage of digital learning environments, which results in improved learning outcomes for students. Despite the robust association between intention and adoption, some researchers contend that intention by itself is not always enough to ensure adoption. For instance, external contexts such as system dependability, technical difficulties and institutional backing can moderate the effect of intention on adoption. Also, Mahande *et al.* (2025) stressed that while students may hold a strong intention toward using AI chatbots, the actual adoption is moderated by motivation, self-efficacy and digital literacy. These constructs point toward the necessity for more studies on the impact of external and psychological factors concerning intention and adoption. Therefore, we suggested that:

H5. Students' intention to use AI chatbots has a significant influence on the adoption of AI chatbots.

2.7 Mediating effect of AI chatbot

The intention to use AI chatbots is a mediator of perceived usefulness, subjective norms, technology literacy and technology simplicity in relation to the adoption of chatbot applications. This implies that users' willingness to adopt chatbots is influenced by their perceptions of these factors through their intention to use the technology. Several elements influence the adoption of AI chatbots in higher education learning including usefulness, subjective norms, tech literacy and technology simplicity (Alzebda and Matar, 2025). Although these elements have a direct impact on students' adoption of AI chatbots, their effects are often mediated by the intention to use AI chatbot applications (Davis, 1989; Polyportis and Pahos, 2025). Perceived usefulness is one of the most important determinants of technology adoption because AI chatbots will make the student's learning experience more fulfilling (Davis, 1989; Polyportis and Pahos, 2025). Students' PU has a significant and positive influence on the intention to use AI chatbots (Tomas and Immerzeel, 2025). However, intention does not always imply adoption. Even if students perceive chatbots as useful, the adoption of the technology may still be inhibited by mistrust, poor system reactions or differing learning styles (Hannigan *et al.*, 2024). This assumes that learners must be willing to accept the integration of chatbots into their learning processes before the actual implementation of chatbot applications (Roca *et al.*, 2024). If educational institutions promote the use of chatbots, students perceive chatbots as an important aid for learning (Ting *et al.*, 2024). While SN can advocate the use of chatbots, students will not readily accept them until there is a self-generated intention to use AI chatbots (Polyportis and Pahos, 2025). External factors do not always assure that people will behave in certain ways. Many studies assume the phenomenon of SN directly influencing adoption (Mustofa *et al.*, 2025; Day *et al.*, 2024) without consideration of the individual's self-intention.

Students with higher technological literacy are more likely to interact with chatbots because they comprehend their features, functions and values (Şahin Kızıl *et al.*, 2025).

However, technological literacy by itself does not guarantee that chatbots will be used. A student who possesses relevant knowledge and skills must also form a personal behavioral intention to use AI chatbots for learning (Polyportis and Pahos, 2025). Intention serves as a significant bridge between digital simplicity and actual adoption; it ensures that students who possess technological skills do not merely wish to use AI chatbots, but purposefully seek to integrate them into their academic life (Song *et al.*, 2025). According to Al-Shafei (2025), when chatbots have simple navigation, clear instructions and intuitive interaction, students are more likely to find them appealing. Although convenience has the potential to promote technology adoption, its effect is more circumstantial. Students are not likely to use a simplistic chatbot interface without first wanting to learn something from it (Xiaoyu *et al.*, 2025). The simpler the interface of the chatbot, the higher the probability that students will wish to interact with and use the chatbot. Most research has dealt with functionality instead of psychological explanations of adoption. Although prior studies emphasize the role of AI chatbot adoption there is limited literature on how students' intention to use AI chatbots as an understudied contextual factor mediates this association. Thus, we proposed the following hypothesis:

H6. Use of AI chatbot intention mediates the effect of perceived usefulness, subjective norm, tech literacy and tech simplicity on the adoption of chatbot applications.

2.8 Moderating effect of tech readiness

Technology readiness (TR) refers to the degree to which an individual is psychologically prepared to adopt new technologies (Leung and Cheung, 2025). In the case of AI chatbots for higher education learning, students' technology readiness can impact their perception of usefulness, subjective norms, technological skills and the automation of AI chatbot applications, which ultimately impacts their usage intentions (Kerimbayev *et al.*, 2025). This study investigates the moderating role of TR in the relationships involving perceived usefulness, subjective norms, tech literacy, tech simplicity and students' intentions to use AI chatbots. Predictive usefulness is a core feature of the TAM, which plays an important role in the adoption of different modern technologies (Ma *et al.*, 2025; Davis, 1989). Students who believe AI chatbots are helpful tend to have a favorable inclination to use them. Nonetheless, the impact of predictive usefulness on intention might be moderated by the students' technological willingness (Teng *et al.*, 2024; Naznen *et al.*, 2023; Alam *et al.*, 2022). Students with high technological willingness are more likely to convert their perception of the usefulness of AI chatbots to the actual intention of usage. Generally, they have positive perceptions of novel technologies and are more willing to use them in learning activities. On the other hand, students with low technology readiness are likely to acknowledge the usefulness of AI chatbots (Abdo-Salloum and Al-Mousawi, 2025).

Although previous research has validated the direct effects of perceived usefulness on technology adoption (Cattaneo *et al.*, 2025), not much attention has been placed on how technology readiness affects this relationship. The effect of technology readiness on subjective norms impacts the intent of users to employ AI chatbots. Students with higher readiness to use technology might be more responsive to their peers and teachers' influence in adopting AI chatbots as they are relatively more interested in new technologies (Abdo-Salloum and Al-Mousawi, 2025). On the other end of the spectrum, students with low readiness to use technology may adopt AI chatbots due to sociocultural pressures, but only reluctantly. It is known how SN affects the adoption of technology (Bhuiyan *et al.*, 2024; Rahman *et al.*, 2022a; Venkatesh *et al.*, 2003), but the effect of tech readiness on this SN and technology adoption relationship has not yet been investigated. On the other hand, low-technology readiness students may not adopt even if there is social pressure to use AI

chatbots (Bouteraa *et al.*, 2024). While numerous studies have shown the impact of subjective norms on technology acceptance (Rahman *et al.*, 2022b; Venkatesh *et al.*, 2003), little has been done to analyze how tech readiness modifies this interrelation.

The interaction of students' levels of technological literacy and technological readiness may either increase or reduce the impact of tech readiness on the intention to use AI chatbots (Abdo-Salloum and Al-Mousawi, 2025; Rahman *et al.*, 2022c). Students who possess high levels of technological literacy and technological readiness are more willing to exhibit a strong intention to use AI chatbots since they are comfortable using technology (Abuadas and Albikawi, 2025). However, students who possess a high technological literacy but have a low technological readiness may be reluctant to use AI chatbots because of their lack of trust or confidence in AI tools (Chanda *et al.*, 2025). Although prior studies emphasize the role of technology literacy in the adoption of new technology, there is limited literature on how technology readiness as an understudied contextual factor moderates this association. Students are more likely to adopt AI chatbots if they perceive them as simple and user-friendly (Can and Nguyen, 2025). Although technology simplicity is an essential factor in technology adoption, few studies have examined how tech readiness moderates its effect (Rahman *et al.*, 2022d). Therefore, we postulated the following hypothesis:

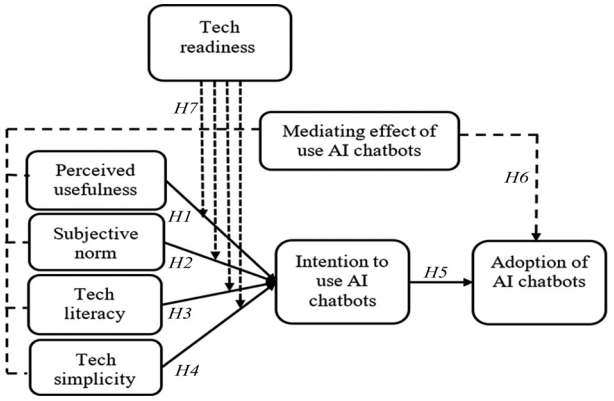
H7. Tech readiness moderates the effect of perceived usefulness, subjective norms, tech literacy and tech simplicity on intention to use AI chatbot applications.

This study aims to fill the existing research gaps by implementing the concepts of TAM and UTAUT as the theoretical framework. Based on the review of the literature and theoretical foundation, Figure 1 illustrates the study's conceptual model whereas Table 1 captures relevant literature and their associated gaps.

The deepening integration of AI chatbots in education has been studied to identify key thematic drivers that explain students' adoption behaviors. Based on the literature findings, three prevailing themes have been identified which lay the groundwork for a conceptual model.

2.8.1 Theme 1: Perceived usefulness and AI chatbots as effective learning tools. Perceived usefulness is one of the key constructs affecting students' use of AI chatbots, and reflects how users think a system improves their productivity (Davis, 1989). For AI chatbots, perceived usefulness is derived from the chatbot's functional and interactive features as well as the efficiency of the task being performed (Shuhaiber *et al.*, 2025). Chen and Anyanwu (2025) reported that the introduction of AI chatbots in Moodle had a significant effect on students' ability to apply metacognitive learning strategies, leading to increased self-regulation of study activities. Abdi *et al.* (2025) brought to light the role of performance expectancy in prospective students' use of ChatGPT, analyzing how students are more likely to adopt AI chatbots if they perceive that using these tools will support their academic achievement. Shuhaiber *et al.* (2025) found that students value ChatGPT in higher education because it provides immediate answers to questions, makes summaries of ideas and facilitates self-learning.

2.8.2 Theme 2: Social influence and the role of subjective norms in AI adoption. Instructors, peers and even institutional policies, which are grouped as subjective beliefs in the TAM and the UTAUT, identified a significant impact in shaping students' propensity toward the use of AI chatbots (Venkatesh *et al.*, 2003). Polyportis and Pahos (2025) noted that students' ChatGPT adoption is affected by the institutional policies and the level of anthropomorphism of the chatbot's interface. Students are likely to adopt the chatbot if they find it to be appealing and "human-like." Abdi *et al.* (2025) examined the impacts of social influence from faculty members and their peers and found that these influences determine



Source(s): Authors' own work

Figure 1. Conceptual model

students' behavioral intentions regarding the use of ChatGPT, particularly in the educational contexts of developing countries such as Somalia. Yusuf *et al.* (2025) created a conceptual framework that argued that pedagogical AI conversational agents are significant within knowledge-sharing cultures in higher education. This pattern implies that peers, educators and even educational institutions must facilitate the use of AI chatbots to improve students' motivation and trust in them.

2.8.3 Theme 3: Technology literacy, simplicity and readiness in AI chatbot adoption. Students understanding and using the features of AI chatbots influence how readily the students adopt these technologies. These factors are divided into two categories such as tech literacy and tech simplicity. Tech literacy refers to students' ability to use advanced AI systems, which affects their readiness and willingness to use chatbots. In contrast, tech simplicity is a user-friendly chatbot interface with minimal complexity that enhances student engagement. Lim and Hwang (2025) found that the activity of the user chatbot and the task complexity influence students to participate in the learning process. Students will use the chatbot functionality more often if their tasks are made easier. According to Gill *et al.* (2024), to carry out a digital shift in the education sector, instructors need to ensure that AI chatbots are readily available to all students regardless of their tech literacy level. Foroughi *et al.* (2025) noted that trust and novelty in design moderate students' adoption of ChatGPT, underscoring the fact that the simplicity and clarity of the chatbot's function and purpose are important for its frequent use. In addition, the adoption of the technology also impacts change readiness. Students who feel ready to use technology are more willing to adopt AI chatbots in their academic work (Zeb *et al.*, 2025; Stohr *et al.*, 2024). Those with negative attitudes toward technology are likely to have difficulties adopting AI, even when they find the technology useful and have sociocultural support.

3. Methods

3.1 Sample and data collection

For this study, data collection aimed to understand students' perceptions of AI chatbot adoption in higher education. A total of 550 students were randomly selected to participate in face-to-face meetings, which took place between March and June 2024. These interactions

Table 1. Gaps of AI chatbot applications in education

Reference	Theoretical context	Antecedents	Dependent/AI chatbot indicators	Findings	Research gaps
Lim and Hwang (2025)	AI-user interaction theory	Proactivity, task types, user characteristics	Chatbot engagement levels	More proactive chatbots enhance user experience	Need for studies on chatbot adaptability and user satisfaction
Yusuf *et al.* (2025)	Constructivist learning theory	Pedagogical AI, conversational agents	AI chatbot framework for higher education	Conceptual model for chatbot-driven learning	Need for empirical validation of the proposed framework
Ganapathy and Kaushal (2025)	Bloom's taxonomy	AI chatbot cognition, anatomy education	Cognitive assessment of ChatGPT vs Gemini	ChatGPT outperforms Gemini in understanding anatomy	Need for studies on AI chatbots in diverse medical fields
Foroughi *et al.* (2025)	Trust theory	Perceived trust, design novelty	ChatGPT adoption in education	Trust moderates adoption behavior	Need for cross-cultural validation of findings
Polyportis and Pahos (2025)	Anthropomorphism theory	Trust, design novelty, institutional policy	Student adoption of ChatGPT	Anthropomorphism increases engagement	Need for exploration of policy-driven chatbot adoption
Zeb *et al.* (2025)	Knowledge sharing theories	AI ethics, academia, libraries	AI impact on knowledge sharing	AI chatbots facilitate learning but raise ethical concerns	Further research on governance and regulatory implications
Chen and Anyanwu (2025)	Metacognitive learning theory	AI-powered chatbots, Moodle-based learning	Academic performance in business education	AI chatbots improve self-regulation and metacognitive learning	Need for comparisons across diverse educational settings
Mohammed *et al.* (2025)	Cognitive learning theories	ChatGPT use in computer education	Academic achievement	ChatGPT enhances students' performance in computer education	Need for studies on long-term impacts and other disciplines
Akpan *et al.* (2025)	Human-AI interaction theories	Conversational AI, generative AI	Human–chatbot interaction in education	AI chatbots improve engagement but lack emotional intelligence	Need for deeper research on long-term human-AI engagement
Abdi *et al.* (2025)	UTAUT	Performance expectancy, effort expectancy, social influence	Behavioral intention to use ChatGPT	Performance expectancy is the strongest predictor	Lack of studies on cultural and institutional differences

(*continued*)

Table 1. Continued

Reference	Theoretical context	Antecedents	Dependent/AI chatbot indicators	Findings	Research gaps
Shuhaiber *et al.* (2025)	TAM	Perceived usefulness, ease of use	Student perspectives on AI chatbots	Students show mixed acceptance; trust and ethics influence use	Further exploration of ethical and psychological effects
Gill *et al.* (2024)	Digital transformation in education	AI-driven learning, ChatGPT effects	AI impact on modern education	AI chatbots transform learning approaches	Need for ethical guidelines on AI use in education
Lin and Yu (2024)	Bibliometric analysis	Research trends in AI chatbots	AI chatbot impact in education	AI chatbot studies are growing in volume	More in-depth studies on effectiveness are needed
Stohr *et al.* (2024)	Gender and educational diversity theories	Gender, academic level, field of study	AI chatbot perceptions among students	Differences in chatbot use by gender and discipline	Need for inclusive and adaptive chatbot designs
Al-Sharafi *et al.* (2023)	Knowledge management theories	Sustainable chatbot use, AI integration	Long-term use of AI chatbots in education	Knowledge management factors influence chatbot sustainability	Need for empirical validation using different AI models
Okonkwo and Ade-Ibijola (2021)	AI in education framework	Chatbot applications	AI chatbot effectiveness in education	Systematic review showing widespread adoption	Need for empirical research on chatbot implementation challenges
Pérez *et al.* (2020)	Systematic literature review	Past chatbot research in education	Evolution of chatbot use in education	Chatbots improve learning engagement	Need for more real-world applications and validation

Source(s): Authors' own work

provided direct insights into their views on using AI chatbots for learning. The primary aim of this study was to collect data with the anonymity of the participants. To reach this goal, no identifying information was gathered. This method permitted learners to express their views without fear of their identity and privacy being overstepped. Anonymity in turn gave impartial and reliable responses, resulting in precise findings. In addition, all data collected was confidential and secured to protect the interests of the participants (Ibrahim *et al.*, 2024). The respondents of the survey were students from private universities located in Kuala Lumpur pursuing their Bachelor, and post-graduate studies. Non-probability random sampling method was used to administer the questionnaires, ensuring that the participants were within the scope of engaged in education at the tertiary level.

Kuala Lumpur was chosen as the location for the research because it represents the most prominent and multifarious educational center in Malaysia. The city boasts numerous private universities of good standing that draw students from all over the country and even abroad. The presence of such heterogeneous students in one city makes Kuala Lumpur a perfect place to study the use of AI chatbots in higher education. Besides, as the capital city of the country, Kuala Lumpur has greater access to digital devices and technology (Kaur and Karim, 2024) which makes it a suitable city for researching the use of AI-based educational software.

In this research, ethics were very important to mitigate bias, ensure transparency and safely measure the rights of participants. Prior to the implementation of the survey, authorization was granted by the research management and innovation center ethics committee. This authorization verified that the research complied with ethical benchmarks for research. To further safety measure participants, an introductory ethical statement was placed at the top of the questionnaire. This statement briefly advised students that they had the option to either accept or decline the invitation to the survey without any pressure. There were no risks or unfavorable outcomes involved in participating in the study. There was no monetary or non-monetary compensation provided to the students for their involvement in the study. Upon reviewing the ethical statement, each student gave written consent before the survey. This confirmed that all respondents comprehended their rights and participated in the study voluntarily. This study followed these ethical protocols, preserving the level of confidentiality, equity and scholarly rigor as well as obtaining valuable information on students' perceptions of AI chatbot integration for learning purposes.

To validate both the reliability and relevance of the questionnaire, we conducted a pretest and a pilot study prior to implementing the full survey. Initially, three specialists in AI chatbot utilization within the educational sector were first consulted to get feedback about the questionnaire. Their assessment highlighted various concerns such as vague wording, overly expansive questions and recurring queries. We improved the questionnaire to increase clarity and refine the effectiveness of the questions. After finalizing the questionnaire, a pilot study was conducted to test its reliability. The Cronbach's alpha results for all variables were above 0.70 (Hair *et al.*, 2019), which ensures that the questionnaire was reliable and valid for data collection as it exhibited internal consistency.

Out of 550 questionnaires distributed, we received 430 responses. To account for bias from non-responses, we systematically performed a missing value analysis. After the omission of four incomplete responses, we were left with 426 valid responses for data analysis, which provides a response rate of 77.45%. This study's target population sample size was determined through the G*Power analysis (Faul *et al.*, 2009), which indicated that at least 109 responses were necessary to achieve 0.80 statistical power. We exceeded this value with the valid responses of 426, allowing for a greater than sufficient sample size for

statistical analysis. These methods ensured that the data collected from the survey was correct, reliable and free from bias, providing strong ground for the study's findings.

3.2 Measurement instrument

The questions in this study were carefully adapted from previous research to evaluate several important constructs. Eight questions from Ayanwale and Molefi (2024), Islam *et al.* (2024) were selected to measure the perceived usefulness and subjective norms. These questions assess how students find AI chatbots and how social influences affect their adoption. Tech literacy was evaluated using four items from Chung *et al.* (2020) and Ait Baha *et al.* (2024). These items measure students' knowledge and skills in using digital technology. Four questions from Ayanwale and Molefi (2024), Razami and Ibrahim (2021) were modified to assess tech readiness. These questions determine students' willingness and preparedness to adopt AI chatbots. Five items were modified from Ayanwale and Molefi (2024) to evaluate the tech simplicity. These questions examine how easily students perceive AI chatbot technology. Additionally, students' intention to use AI chatbot applications was assessed with four items adapted from Ayanwale and Molefi (2024), Tian *et al.* (2024) and Rahman *et al.* (2025), whereas five reformed questions from Maheshwari (2023), Tian *et al.* (2024), Al-Abdullatif and Alsubaie, 2024) and Schei *et al.* (2024) were used to measure the adoption of AI chatbots. These questions analyze students' willingness to use AI chatbots for learning and assess the extent to which students have adopted or plan to adopt AI chatbots. By using validated questions from previous studies, this research ensures that the survey is reliable and effectively captures students' perceptions of AI chatbot adoption.

This research used a five-point Likert scale for each survey item with categories from "strongly disagree" to "strongly agree". This approach allows for an orderly collection of the respondents' opinions. The use of a Likert scale improves clarity and consistency, balanced responses and accuracy in evaluation. Respondents can comprehend and address each statement, which minimizes misunderstanding. The scale offers a neutral midpoint which stops extreme responses from taking place as well as helps mitigate bias. It aids in the assessment of strategic issues regarding the use of AI chatbots intention and the adoption of AI chatbots for effective learning purposes.

3.3 Common method bias

In assessing for common method bias (CMB), we used the variance inflation factor (VIF). As Kock (2017) and Podsakoff *et al.* (2012) described if the VIF value stands at 3.3 or above, CMB exists, implying the results are susceptible to error. However, if all values stay below 3.3, the model does not contain CMB. In this instance, all VIF values were below 3.3 (see Table 3), which illustrates that CMB is not an issue in the data set. In addition, we examined skewness and kurtosis values to check for the normality of the data. Sheridan and Coakes (2011) state that a normal distribution is established if the skewness values range from −1.5 and 1.5. Normality may also be confirmed if the kurtosis values also range from −2.0–2.0. The skewness and kurtosis values in this study are achieved within these ranges, hence the data is normal for the purpose of analysis.

3.4 Data analysis technique

For this study, we analyzed the information using IBM SPSS 24.0 and SmartPLS 4.0. Each software assisted with the individual aspects of the research. IBM SPSS 24.0 was used to analyze demographic variables such as age, sex and level of education. This analysis served to describe the participants and demographic information regarding the sample. IBM SPSS 24.0 was used to analyze demographic variables such as age, sex and level of education. This

analysis served to describe the participants and demographic information regarding the sample. SmartPLS 4.0 was used for testing relationships between variables, which is based on partial least squares structural equation modeling (PLS-SEM). This technique facilitates predictive modeling and complex interrelationships among many variables. The study was conducted in two stages (Hair *et al.*, 2019; Chin, 2010). Stage one is measurement model analysis. This stage evaluated how dependable/reliable and valid the measurement scales were checked the data for suitable conditions to test the proposed hypotheses. Stage two is structural model analysis. In this step, a series of hypotheses were tested using the bootstrapping procedure. Bootstrapping computes the statistical significance of a relationship by estimating the sample size from an original data set. Using these highly sophisticated statistical methods, the study ensures reliable and accurate findings in predicting the interconnections between important variables of AI chatbot adoption.

4. Results and discussion

4.1 Demographic information

The findings revealed that female respondents were 61.7% and male respondents were 38.3%. The largest age group is 26–30 years old with 36.3%, followed by 31–35 years old with 23.4%, above 36 years old with 23.3% and 20–25 years of 17%. The survey participants came from various academic disciplines with 13.4% from science and technology, 33.0% from social science, 16.0% from engineering and 37.6% from health science. This diverse representation helps provide a well-rounded understanding of AI chatbot adoption across different fields of study. Regarding AI chatbot usage, 42.3% of students used AI chatbots for less than 5 h daily, 28.2% used them for 5–10 h daily and 29.5% used them depending on the situation. For the academic year distribution, 25.8% of respondents were in year one, 28.7% in year two, 23.0% in year three and 22.5% in year four. This distribution ensures a balanced representation of students at different stages of their academic journey, providing insights into AI chatbot adoption across various levels of education. The majority of the respondents (57%) are pursuing bachelor degrees, followed by 29.8% enrolled in master and 13.2% pursuing PhD The details of the demographic profile are described in Table 2.

4.2 Measurement model analysis

The measurement model was assessed to check for the reliability and validity of the constructs in this study. The factor loadings were between 0.677 and 0.925 (Figure 2) and all were above the suggested minimum cut-off of 0.70 (Hair *et al.*, 2019) except the item "I understand how AI chatbots work and how they assist in learning (tl1)", which supports that the items in the study represent their constructs. Hence, we dropped the item "I feel social pressure to adopt AI chatbots in my daily activities (sn1)" from subjective norms due to the lower factor loading of 0.57. To establish convergent validity, composite reliability (CR) as well as average variance extracted (AVE) need to be greater than or equal to 0.70 (Hair *et al.*, 2019). The AVE values were also between 0.545 and 0.802 which shows that there is an adequate amount of variance captured by the constructs. The values of CR ranged from 0.827–0.942 (Table 3) which ensured the internal consistency and reliability of the measurement model. The values of the variance inflation factor (VIF) ranged from 1.301–2.796 all lower than the critical cut-off of 3.3 (Kock, 2017). This proves that multicollinearity is not an issue for this study. This analysis demonstrates that the measurement model exhibits proper statistical validity confirming that the constructs will measure accurately for further data analysis.

After confirming convergent validity, the next stage was checking for discriminant validity, which ensures that the constructs are different and not similar. Fornell and Larcker (1981) explained that the discriminant validity is confirmed when the square root of the AVE

Table 2. Demographic characteristics of the participants

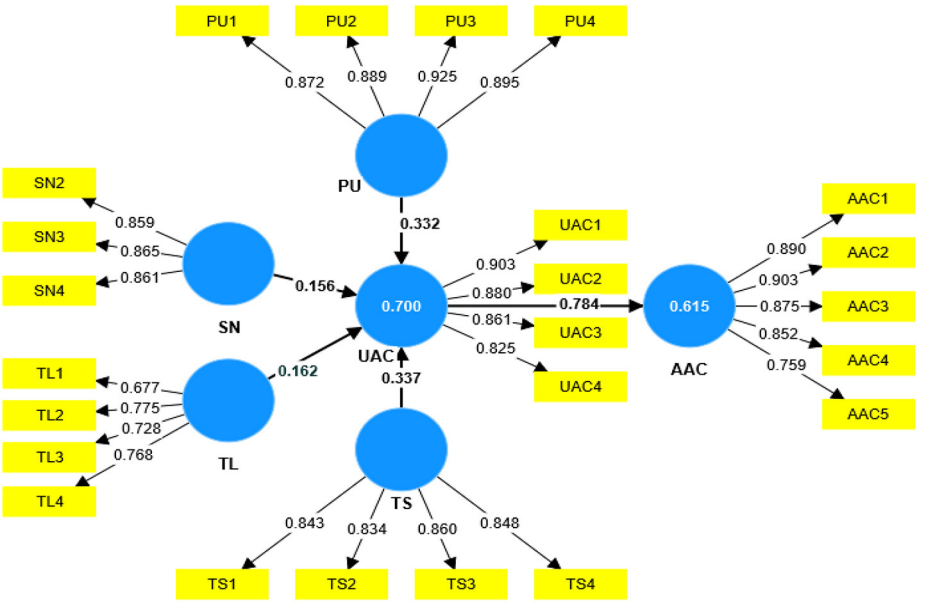
Characteristics	Frequency	%
Gender		
Male	167	38.3
Female	259	61.7
Age group		
20–25 years	78	17.0
26–30 years	145	36.3
31–35 years	103	23.4
Above 36 years	100	23.3
Faculty		
Science and technology	57	13.4
Social science	141	33.0
Engineering	68	16.0
Health science	160	37.6
Using the AI chatbots		
Below 5 h daily	180	42.3
5–10 h daily	121	28.2
Depend on situation	125	29.5
Academic year		
Year 1	111	25.8
Year 2	120	28.7
Year 3	101	23.0
Year 4	94	22.5
Education level		
Bachelor	243	57.0
Master	127	29.8
PhD	56	13.2

Source(s): Authors' own work

of each construct is greater than its correlations with other latent constructs. This implies that each construct must have stronger relationships with its indicators than with those of the other constructs. Franke and Sarstedt (2019) proposed that an HTMT value below 0.90 is sufficient to confirm discriminant validity. In this analysis, all values of HTMT (Table 4) were lower than 0.90, which shows that the constructs are not statistically similar to one another. These findings provide sufficient justification that the measurement model meets the discriminant validity requirements, which ensures that the constructs are not overlapping and are precisely defined.

4.3 Structural model analysis

The structural model was examined for hypothesis testing using the bootstrapping method. The results revealed that perceived usefulness has a highly significant and positive influence on students' intention to use AI chatbots ($\beta = 0.332$, $t = 6551$, $p < 0.01$). Subjective norms ($\beta = 0.153$, $t = 3.412$, $p < 0.01$), tech literacy ($\beta = 0.162$, $t = 3.233$, $p < 0.01$) and tech simplicity ($\beta = 0.337$, $t = 5.942$, $p < 0.01$) have highly significant and positive impacts on intention to use AI chatbots. Therefore, hypothesis *H1*, *H2*, *H3* and *H4* were accepted. Students' intention to use AI chatbots has a highly significant and positive impact on the



Source(s): Authors' own work

Figure 2. Measurement model

adoption of AI chatbots in higher education learning (β = 0.784, t = 28.956, p < 0.01), as a result, *H5* is accepted (Figure 3, Table 5). The results also indicated that the intention to use of AI chatbot has a partial mediation in the effect of perceived usefulness (β = 0.260, t = 5.922, p < 0.01), subjective norms (β = 0.122, t = 3.416, p < 0.01), tech literacy (β = 0.127, t = 3.200, p < 0.01) and tech simplicity (β = 0.264, t = 6.114, p < 0.01) on the adoption of AI chatbot applications. Hence, the results showed that tech readiness moderates the effect of perceived usefulness, and tech simplicity on students' intention to use AI chatbot applications (Table 5, Figure 4). However, tech readiness does not moderate the effect of subjective norms and tech literacy on the intention to use AI chatbot applications. This suggests that external influences (such as peer pressure or social influence) and students' basic digital skills are not significantly altered by their overall readiness for technology. These results emphasize the importance of perceived usefulness, technological literacy, ease of technology and subjective norms in determining students' intention to use AI chatbots and adoption. Furthermore, technological readiness amplifies some considerations (i.e. usefulness and simplicity) but remains neutral to the effects of social influences and digital competencies. These findings enhance the comprehension of the engagement patterns of students with AI-powered learning systems in higher education.

The blindfolding procedures were used in the current investigation to assess the structural model's predictive relevance (Q^2) and effect size (f^2). The outcomes showed that the model had a good ability to predict the variables. For example, Q^2 values for the use of AI chatbots and adoption of AI chatbot applications were determined to be 0.455 and 0.465, respectively. Similarly, R^2 values for students' intention to use AI chatbots and adoption of AI chatbot

Table 3. Convergent validity

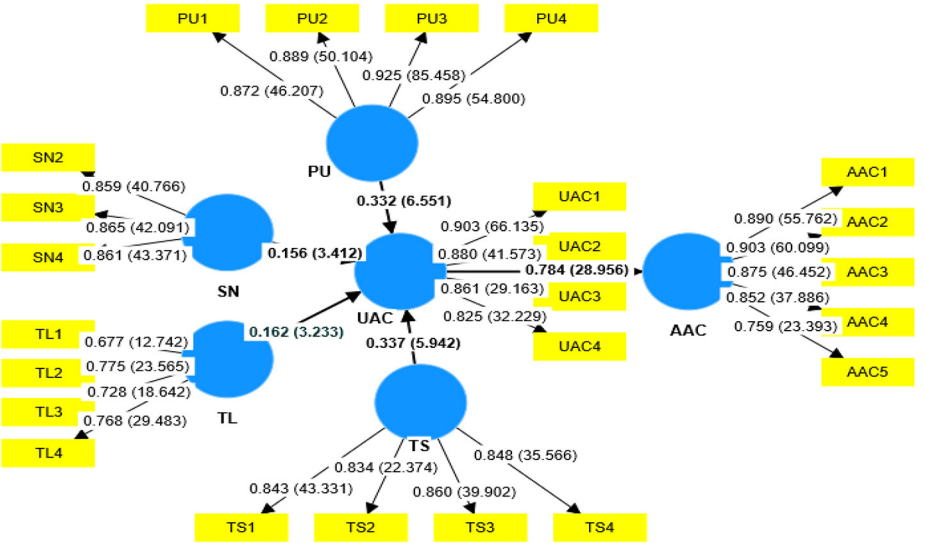
Characteristics	Skewness	Kurtosis	VIF	FL	CA	CR	AVE
Perceived usefulness					0.918	0.942	0.802
Using AI chatbots improves my learning performance (pu1)	−0.305	0.322	1.967	0.872			
AI chatbots help me complete my learning tasks more efficiently (pu2)	−0.440	0.313	2.102	0.889			
Using AI chatbots enhances my understanding of course materials (pu3)	−0.316	−0.243	2.032	0.925			
AI chatbots save me time when searching for learning resources (pu4)	0.254	0.346	1.821	0.895			
Subjective norm					0.827	0.896	0.743
My classmates encouraged me to use AI chatbots for my studies (sn2)	−0.605	0.403	1.799	0.859			
My lecturers recommend using AI chatbots for learning (sn3)	−0.363	−0.491	2.075	0.865			
Most students in my university use AI chatbots for academic purposes (sn4)	−0.345	−0.237	1.845	0.861			
Tech literacy					0.726	0.827	0.545
I understand how AI chatbots assist in learning (tl1)	−0.503	−0.321	1.327	0.677			
I frequently use digital platforms for my studies (tl2)	−0.324	−0.212	1.407	0.775			
I feel comfortable using AI chatbot applications for academic purposes (tl3)	−0.407	0.342	1.301	0.728			
I can quickly learn how to use new technology without assistance (tl4)	−0.317	0.289	2.073	0.768			
Tech simplicity					0.868	0.910	0.716
AI chatbots are easy to use for learning (ts1)	−0.424	0.548	1.944	0.843			
I can interact with AI chatbots without needing technical knowledge (ts2)	−0.304	0.536	1.867	0.834			
AI chatbots provide clear instructions that are easy to follow (ts3)	−0.365	0.473	1.835	0.860			
I do not need extensive training to use AI chatbots effectively (ts4)	−0.337	0.532	2.615	0.848			
Use of AI chatbots					0.891	0.924	0.753
I intend to use AI chatbots for my learning activities (uac1)	−0.431	0.203	1.882	0.903			
I am likely to use AI chatbots to support my studies (uac2)	−0.513	0.375	2.305	0.880			
I plan to use AI chatbots regularly for academic purposes (uac3)	−0.315	0.428	2.112	0.861			
I will recommend AI chatbots to my classmates for learning assistance (uac4)	−0.313	0.517	1.738	0.825			
Adoption of AI chatbots					0.909	0.932	0.735
I have started using AI chatbots for my learning activities (aac1)	−0.322	−0.164	2.750	0.890			
I frequently use AI chatbots to assist with my studies (aac2)	−0.284	−0.425	2.375	0.903			
I prefer AI chatbots over traditional learning support methods (aac3)	−0.242	−0.538	2.796	0.875			
I feel comfortable using AI chatbots for academic tasks (aac4)	−0.253	−0.512	2.625	0.852			
I believe AI chatbots improve my overall experience (aac5)	−0.312	0.578	2.045	0.759			

Note(s): Variance inflation factor (VIF), Factor loading (FL), Cronbach's alpha (CA), Composite reliability (CR), Average variance extracted (AVE)
Source(s): Authors' own work

Table 4. Discriminant validity

Fornell–Larcker criterion	AAC	PU	SN	TL	TS	UAC
Adoption of AI chatbot (AAC)	0.857					
Perceived usefulness (PU)	0.862	0.896				
Subjective norms (SN)	0.526	0.532	0.862			
Tech literacy (TL)	0.607	0.623	0.574	0.738		
Tech simplicity (TS)	0.638	0.623	0.588	0.673	0.846	
Use of AI chatbots (UAC)	0.784	0.726	0.623	0.685	0.744	0.868
Heterotrait–monotrait (HTMT)						
Adoption of AI chatbot (AAC)	–					
Perceived usefulness (PU)	0.643					
Subjective norms (SN)	0.600	0.606				
Tech literacy (TL)	0.718	0.731	0.737			
Tech simplicity (TS)	0.713	0.692	0.687	0.829		
Use of AI chatbots (UAC)	0.867	0.801	0.721	0.825	0.842	–

Note(s): Adoption of AI chatbot (AAC), Perceived usefulness (PU), Subjective norms (SN), Tech literacy (TL), Tech simplicity (TS), Use of AI chatbots (UAC)
Source(s): Authors' own work



Source(s): Authors' own work

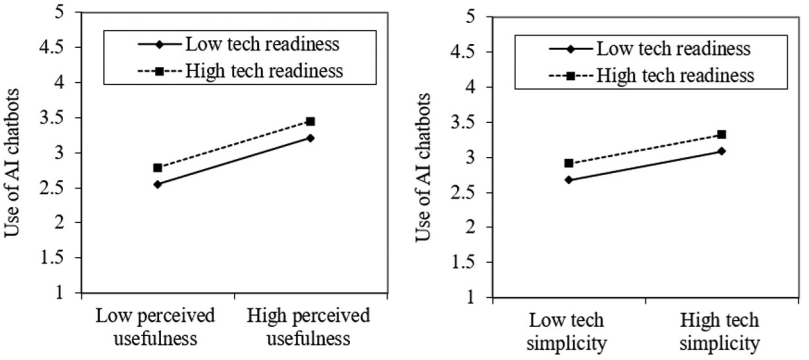
Figure 3. Structural model

applications were found to be 0.700 and 0.615 respectively. To examine the strength of the association between the variables, effect size (f^2) values were also evaluated. According to Cohen (1988), a f^2 value of 0.35 denotes a significant impact size, whereas a value of 0.02 denotes a modest effect size. The results showed that perceived usefulness (0.146), perceived subjective norms (0.255), tech literacy (0.212), tech simplicity (0.345) and use of AI chatbots (0.427) all had significant f^2 values.

Table 5. Path coefficient

HP	Relationship	β	SD	*T*-values	R^2	Q^2	f^2	Comment
H1	PU → UAC	0.332	0.051	6.551**			0.146	Significant
H2	SN → UAC	0.153	0.046	3.412**			0.255	Significant
H3	TL → UAC	0.162	0.050	3.233**			0.212	Significant
H4	TS → UAC	0.337	0.057	5.942**	0.700	0.455	0.345	Significant
H5	UAC → AAC	0.784	0.027	28.956**	0.615	0.465	0.427	Significant
Mediating effect: Intention to use AI chatbots								
H6a	PU → UAC → AAC	0.260	0.044	5.922**				Partial mediation
H6b	SN → UAC → AAC	0.122	0.036	3.416**				Partial mediation
H6c	TL → UAC → AAC	0.127	0.040	3.200**				Partial mediation
H6d	TS → UAC → AAC	0.264	0.043	6.114**				Partial mediation
Moderating effect: Tech readiness								
H7a	TR x PU →UAC	0.329	0.065	5.061**				Significant
H7b	TR x SN → UAC	0.031	0.087	0.356				Not significant
H7c	TR x TL → UAC	0.055	0.093	0.591				Not significant
H7e	TR x TS → UAC	0.204	0.058	3.517**				Significant

Note(s): Perceived usefulness (PU), Subjective norms (SN), Tech literacy (TL), Tech simplicity (TS), Use of AI chatbots (UAC), Adoption of AI chatbot (AAC), Hypothesis (HP), Beta value (β), Standard deviation (SD). Significant level at *p< 0.05, **p< 0.01
Source(s): Authors' own work



Source(s): Authors' own work

Figure 4. Moderating effect of tech readiness

4.4 Importance-Performance matrix analysis

The Importance-Performance Matrix Analysis (IPMA) is a valuable tool for identifying key factors that drive performance in a research model. It helps researchers determine which constructs have the most significant impact on the desired outcome while considering their performance levels (Hair *et al.*, 2019). The (IPMA identified that tech simplicity appeared as the crucial factor for the performance of AI chatbot applications (0.484; 67.530). This suggests that students who possess strong technological skills are more likely to perceive AI chatbots as simple and easy to use, reinforcing the importance of tech simplicity in AI adoption. The next

most significant factor was perceived usefulness (0.260; 63.208). These findings align with prior studies (Davis, 1989; Venkatesh and Davis, 2000), which assert that users are more willing to embrace new technology if they perceive it as useful in improving their learning. The importance score of AI chatbots equated to 0.372 while the performance score was 62.971. This shows that the prior use of AI tools by students affects their attitude toward the adoption of chatbots. These findings support the technology acceptance literature, which demonstrates that prior exposure to a system enhances users' trust and readiness to use it in educational tasks (Gefen *et al.*, 2003). In addition, the adoption of chatbots was also influenced by tech literacy (0.127; 60.744) and subjective norms (0.122; 59.780) but to a lower degree (Table 6 and Figure 5). The effect of tech literacy is somewhat consistent with prior research (Chung *et al.*, 2020), showing that having basic digital skills facilitates the adoption of chatbots, but it is not the main reason. On the other hand, subjective norms had the least effect, which means social influences are not as strong as one's capabilities and perceived value for using chatbots for learning (Fishbein and Ajzen, 1975; Venkatesh *et al.*, 2003).

5. Discussion

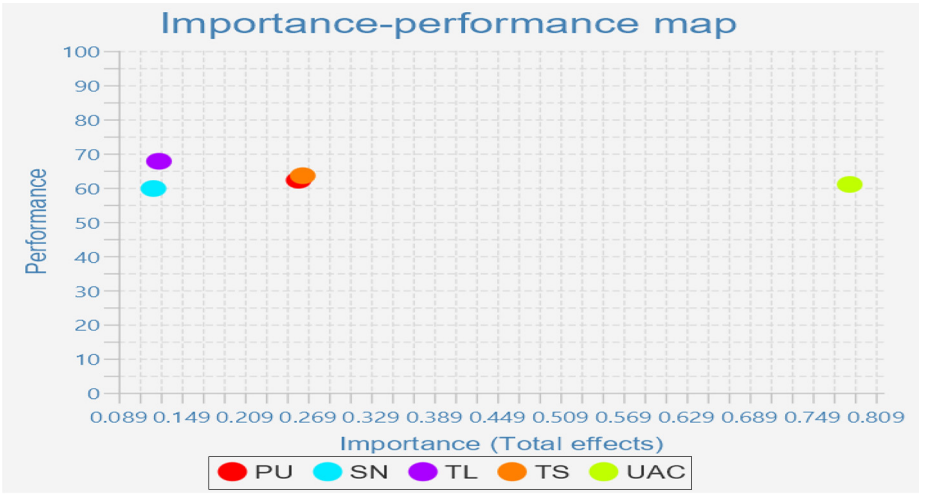
The results showed that perceived usefulness has a significant and positive impact on students' intention to use AI chatbots (*H1*). This is supported by existing research by Islam *et al.* (2024) where students' perceptions regarding the efficiency of AI chatbots as learning tools impacted how actively students used these technologies leading to improved learning outcomes. Similarly, Karaahmetoğlu *et al.* (2023) emphasized that learners' acceptance of AI chatbots for educational purposes is affected largely by the perceived usefulness of the AI chatbots. The more students believe that chatbots can leverage their learning levels, the more they are willing to adopt the chatbots in their learning activities. In addition, AI chatbots have proven to be useful secondary aids for students with disabilities because they provide personal help tailored to the specific learning needs of individuals (Habibi *et al.*, 2025). The combination of these findings highlights the importance of perceived usefulness in the adoption and proper use of AI chatbots for educational purposes. The more students recognize the benefits of these technologies, the more their application in educational settings is bound to increase, leading to better learning outcomes.

The results also found that subjective norms have a significant and positive influence on the intention to use AI chatbot applications (*H2*). This result is associated with earlier studies by Islam *et al.* (2024) who claimed that social support from coworkers, teachers and the organizational culture impacts the adoption of AI chatbots by learners. If students believe that their classmates or teachers support the use of AI chatbots, there is a higher chance that the students will start employing these tools in their learning activities. Al-Abdullatif and Alsubaie (2024) noted that social or subjective norms impact students' decisions regarding

Table 6. Performance and total effect assessment

Target construct: adoption of AI chatbots		
Variables	Total effect	Performance
Perceived usefulness	0.260	63.208
Subjective norms	0.122	59.780
Tech literacy	0.127	60.744
Tech simplicity	0.484	67.530
Use of AI chatbots	0.372	62.971

Source(s): Authors' own work



Source(s): Authors' own work

Figure 5. Importance-performance map (adoption of AI chatbots)

the acceptance of technologies in higher education learning. In their study, they observed that students perceive greater social and institutional encouragement to interact with the AI chatbots when there is a high promotional activity regarding them within the academic environment. Further, Maheshwari (2023) suggested that subjective norms stemming from peers or faculty members are stronger in formal educational settings where students are scoped to a particular instructional technology. These findings imply that the adoption of AI chatbots among students can be improved by creating an encouraging environment where the use of chatbots is actively promoted and supported. By treating AI chatbot applications as vital learning aids, institutions are likely to improve the attitude and involvement of students toward these technologies.

Tech literacy has a significant and positive relationship with students' intention to use AI chatbot applications (*H3*) because students who are more adept with technology tend to have more positive intentions in higher education learning adoption. This finding supports Ait Baha *et al.* (2024), who reported that people with more advanced technological skills are more readily willing to use AI-powered chatbots. Their research proposed that users who are proficient with technology can take advantage of AI chatbots more than people with low computer literacy. Nguyen *et al.* (2022) noted that users with deep technical skills have a relatively easier time dealing with the user interfaces and functions of chatbots. Their understanding and utilization of chatbot features motivate them to incorporate these tools more easily into the learning process. This suggests that technologically savvy students are more likely to accept the use of AI chatbots, which results in an intention to use the application of technology in education. These results suggest the necessity to raise the tech proficiency of students, thus easing the integration of AI-powered chatbots in remote learning. This can be achieved through the provision of digital competencies workshops and the incorporation of chatbots in coursework to build comfort and trust in AI educational technologies.

Tech simplicity has a significant and positive influence on students' intention to use AI chatbot applications (*H4*). This finding is relevant to Ayanwale and Molefi (2024) who

indicate that students tend to adopt AI chatbots with friendly graphical user interfaces for learning. It was noted that simplification in chatbot interaction helps to remove obstacles to participation and captivates the user's interest which results in increased utilization of the chatbot. Ma *et al.* (2025) noted that users are likely to favor AI chatbots that are easy to understand and use, suggesting that technology adoption is driven by the perceived simplicity of the device. Their study noted that students' confidence in using chatbots for educational purposes increased when the chatbots offered reasonable responsive interactions, provided clear instructions and had intuitive features. Also, Tian *et al.* (2024) showed that students are more willing to use AI educational tools if their engagement with the tool is effortless. They reasoned that the students' willingness to integrate the use of chatbots into their daily learning activities is enhanced when the students view the chatbots as easily accessible information, query and academic discussion facilitators. These findings underscore the need for AI chatbots to be designed with the end user in mind, and with a primary focus placed on ease of use enhanced adoption, and meaningful engagement with the technology in higher education learning contexts.

AI chatbot capability has a highly significant and positive relationship with the adoption of AI chatbots (*H5*). This observation is consistent with Maheshwari (2023) who indicated that AI-powered chatbots equipped with natural language processing and adaptive learning features are more likely to be adopted by users in educational contexts. Chatbots with higher capabilities tend to enjoy a higher adoption rate as students are willing to use them for academic support and interactive learning activities. Similarly, Tian *et al.* (2024) indicated that the degree of capability of an AI chatbot remains a significant determinant of students' engagement with learning resources. Their research showed that students' willingness to use learning tools increased substantially when chatbots offered correct answers, proper guidance and easy assistance with educational technologies. Furthermore, Al-Abdullatif and Alsubaie (2024) pointed out that the chatbot's capabilities affect adoption through strengthened perceived trust and dependability. Students are more likely to adopt AI chatbots when they perceive them to be smart, helpful and attentive to their educational requirements. Their results indicate that enhancing the functional capabilities of the chatbot, including but not limited to, real-time feedback, appropriate contextualization and responsive learning suggestions, will increase adoption in digital learning environments.

The intention to use AI chatbot mediates the effect of perceived usefulness, subjective norms, tech literacy and tech simplicity on the adoption of AI chatbots (*H6*). The finding is related to Islam *et al.* (2024) who found that perceived effectiveness serves as an important determinant in students' adoption of AI chatbots through their intentions to use them. In cases where students believe AI chatbots assist with improving learning outcomes, there are higher chances that they will develop a strong intention to use the chatbots which will eventually be translated to higher adoption rates. Moreover, the mediating role of intention to use AI chatbots is consistent with findings from Karaahmetoğlu *et al.* (2023), where they noted that students are highly influenced by subjective norms with technology adoption. Social influences from peers, instructors or institutional encouragement create a sense of obligation or motivation to use AI chatbots, which strengthens their intention for usage and adoption. Perceived usefulness and ease of use as major predictors of technology adoption were recognized by the TAM model. Davis *et al.* (1989) stated that technology will be adopted when the prospective users perceive that it will be beneficial, and simple to use in their learning processes. In addition, Ait Baha *et al.* (2024) revealed that tech literacy increases the use intention of AI chatbots, as students who are more digitally skilled are more at ease interacting with and comprehending the functions of the chatbot's interface. Likewise, Nguyen *et al.* (2022) proposed that the perception of ease in using chatbots

increases the likelihood of students interacting with these tools, thus enhancing their use intention.

The results revealed that tech readiness moderates the effect of perceived usefulness, and tech simplicity on the intention to use AI chatbot applications (*H7*). This implies that students who feel more ready and at ease with technology are more likely to view AI chatbots as beneficial resources, which increases their intention to use them for online learning. Similarly, tech readiness strengthens the impact of tech simplicity on students' intent to use AI chatbots. Students who are technologically ready consider AI chatbots to be user-friendly, which makes them more willing to use such chatbots in their learning. In addition, when students are well-equipped, technically supported and possess the required resources, it appears that their perceptions of the usefulness of AI technology and their tech competencies are greater in their adoption of AI chatbots (Chanda *et al.*, 2025). This is in line with prior studies that assert that the adoption of technology in education is determined by external support (He *et al.*, 2023; Timotheou *et al.*, 2023). Users' perception of usefulness and simplicity in technology may be supplemented by facilitating conditions regarding the AI chatbots. Liesa-Orús *et al.* (2023) determined that technological simplicity such as institutional support may increase the impact of perceived usefulness on technology adoption. Nevertheless, the findings indicate that tech readiness does not moderate the influence of subjective norms and tech literacy on the intention to use AI chatbots. This suggests that students' level of technological readiness does not significantly change the extent to which social factors or their level of technology knowledge inform how they decide to adopt the chatbots. These results support Ait Baha *et al.* (2024) and Islam *et al.* (2024) who highlighted the role of perceived ease of use and individual preparedness in technology adoption.

5.1 Theoretical contribution

This study offers significant theoretical implications by contributing to the understanding of AI chatbot adoption in learning environments and integrating recent research findings into this field. Extension of the TAM, our findings support the fundamental components of TAM, namely, perceived usefulness and perceived ease of use or tech simplicity as it relates to AI-driven learning technologies. This is consistent with Ibrahim *et al.* (2024) who corroborated an extended TAM with the incorporation of personality traits and an AI mindset, underscoring the importance of perceived usefulness in having a positive intention toward AI integration. The findings provide new insight into the impact of perceived usefulness, subjective norms, technology literacy and technology simplicity within an AI chatbot educational environment. This study has a different focus from conventional models of technology adoption, which typically focus on usability and the ease of using the technology (Davis, 1989). It underscores the relevance of both internal and external resource factors, such as the level of technology readiness and the use of AI chatbots, as moderation and mediation variables of adoption and learning outcomes.

The study builds on the basic structure of TAM by showing how students' perceived usefulness of AI chatbots has an impact on their intention to use them, which supports the findings of Islam *et al.* (2024) and Karaahmetoğlu *et al.* (2023). These analyses illustrate that students are more willing to interact with AI learning platforms when they consider the chatbots to be useful for enhancing their learning outcomes. The study also provides a counterpoint to prior views regarding technology acceptance by stressing the role of technology literacy and ease of use as major determinants of students' interactions with AI chatbots. One of the contributions of this study is the re-evaluation of subjective norms regarding the adoption of AI chatbots. The findings suggest that subjective norms play a

significant role. Nguyen *et al.* (2022) stated that students' peers and instructors influence their perception of AI technologies. However, this study contradicts earlier TAM models that downplayed social influence in learning technology adoption, suggesting that in AI-driven environments, social norms still shape students' intentions but are less dominant than functionality and usability concerns.

Besides, the gap in technological literacy emerged as an important barrier to using AI Chatbots by students, which is in line with Ait Baha *et al.* (2024) who stated that students who are more technologically skilled are more responsive toward the use of AI teaching tools. This indicates that students with higher levels of technological skills are more likely to appreciate AI chatbots and consequently are more open toward their application in the educational system. This research incorporates technology readiness as a principal moderating variable that affects the association of perceived usefulness and the simplicity of technology on the intention of the students to use AI chatbots. The results are consistent with Ibrahim *et al.* (2024) who pointed out that students' willingness to interact with AI chatbots influences the adoption rate. Furthermore, this research shows that the intention to use AI chatbots takes control of the relationship between perceived usefulness, subjective norms, tech literacy, simplicity of technology and adoption of chatbot applications. These findings contribute to the existing literature on the adoption of AI in higher education from the point of view of the readiness of the users and the features of the technology and the social context that provide integration of external and internal factors.

5.2 Practical contribution

This study provides diverse practical and policy contributions for educators, institutions, policymakers and even developers working with AI technology. Identifying the most significant drivers of AI chatbot adoption enables stakeholders to devise plans that improve technology integration, student participation and educational results. The results of the study have highlighted the impacts of perceived usefulness, tech literacy and tech simplicity in the adoption of AI chatbots. Developers and education technology providers concentrate more on the AI interface design than the fundamental educational principles that underpin their use. To meet students' learning expectations, AI chatbots need to incorporate adaptive feedback, personalization and interactive guidance features. Considering the possible impact of cultural diversity, AI chatbots must be adapted to suit the local languages and region's learning styles. For the effective integration of AI chatbots into the educational curriculum, universities and educational institutions should ensure that students are AI-driven tools ready. All educational institutions should introduce AI and digital skill training to raise the confidence and competence of their students when interacting with AI-based educational tools. Policymakers need to support a mixed approach to learning where teaching is enriched using AI chatbots to improve learners' engagement and outcomes. Educators should be trained to use AI chatbots effectively in instruction by using AI to provide personalized learning and teaching assistance.

The study highlights the importance of technological readiness as a prerequisite for moderating the adoption of AI Chatbot technology, noting that institutions must build appropriate infrastructure and provide adequate digital access for AI-based learning. Universities need to improve their technology infrastructure by providing high-speed internet, AI-enabled learning systems and cybersecurity systems to facilitate chatbot-based learning. Institutions should provide instructional sessions, and facilitate live-on-campus tech help, and other support services so that students can use AI chatbots effectively within the learning environment. Policymakers and government authorities should create budgetary allocations, grants and other incentives for the adoption of AI to assist institutions in

incorporating AI-based learning tools into their educational systems. Governments must create policies outlining responsible AI practices in education that encompass bias and ethics as well as data privacy concerning AI Chatbots. Institutions should collaborate with the ministries of education and other regulatory agencies to formulate AI policies for educational processes that maintain quality assurance systems and academic standards. Universities and policymakers should foster international partnerships to exchange best practices, research findings and technological advancements in AI-driven learning.

5.3 Limitations and future research direction

This research integrates the TAM with other concepts by offering a detailed explanation of the phenomenon of technology adoption in AI chatbot educational environments. This study extends the literature by integrating the perception of an AI chatbot's usefulness and ease of interaction with other determinants such as technology readiness, subjective norms, tech literacy and tech simplicity which have not been integrated into TAM-based models. The findings indicate that to enhance students' interactions with AI chatbots, both external and internal sources need to be examined. Emotional engagement and trust with AI chatbots, the sociocultural gap in AI acceptance, as well as, the transformation of social influence are some of the front-running areas of research. The effectiveness of AI chatbots in learning is also influenced by students' emotional attachment to the technology. Future research should explore the role of trust in engagement with AI learning tools. This study was conducted with university students in Malaysia. While the findings are useful, sociocultural issues may impact AI acceptance. The different perceptions of AI technologies by students from various regions require further investigation into how sociocultural background shapes the adoption of AI chatbots for learning.

Although this study contributes to the literature in several metrics, some areas need improvement. The sample used comprised students of different Malaysian universities, who might not capture the views of students from different cultural and institutional educational settings. Attentiveness must be exercised when applying these results to other student populations due to the existing differences in AI chatbot adoption the region's technological infrastructure, policy frameworks and digital literacy levels. Also, though this study examines students' perception of AI chatbot adoption, this perception changes over time and remains undetermined by any longitudinal studies. Other researchers may investigate how students' willingness to use AI technologies in learning changes after being exposed to such technologies in education for a longer period. This study fills a critical knowledge gap regarding students' adoption of AI chatbots for learning effectiveness. By examining motivating and inhibiting factors such as perceived usefulness, subjective norms, tech literacy, tech simplicity and tech readiness, the study provides a foundation for future studies.

Further studies can analyze if the patterns observed in this study apply to different student populations and teaching settings, as well as to the differing levels of technology available. Since students vary in their tech literacy and readiness, further research should analyze how these differences affect the adoption of AI chatbots. Other constructs such as self-efficacy, motivation and digital trust could further explain why certain students are more inclined to adopt AI chatbots for learning tools. The integration of TAM with moderating and mediation variables adds an additional layer of construct complexity regarding the adoption of AI chatbots for educational purposes. By modifying the social influence, technology literacy and perceived usefulness, this study goes further to propose new directions for future theorization and real-world application. Understanding the impacts of cross-culture, intention and the affective aspects, AI chatbots will deepen the knowledge of acceptance of technology in education.

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Corresponding author

Muhammad Khalilur Rahman can be contacted at: mohdkhalilur@gmail.com

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